

SAMSUNG PRISM GENAI HACKATHON · 3RD EDITION · 2026-27

THEME 2

Smart Guided Troubleshooting Engine

From vague Galaxy complaints to validated, ordered troubleshooting plans —
with one-tap Settings deeplinks.

Team – Agentic Alchemists

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100% DEEPLINK ACCURACY · ~40 MS CACHED ANSWERS · \$0.00 / QUERY

Troubleshooting shouldn't feel like a maze

01

"screen flickers", "wifi keeps dropping"

Users speak in symptoms.
Support docs speak in jargon.

02

Hundreds of Settings screens

Users get lost four taps deep,
following text instructions.

03

Chatbots invent answers

Fake menu paths. Wrong fixes.
Zero trust.

Self-serve has to be right the first time — not approximately right.

Vague complaint in. Validated plan out.

INPUT

**"My screen keeps
flickering..."**

+ SIIIS support article



ENGINE

**Enrich → ground
→ retrieve → validate**

Deterministic pipeline



OUTPUT

**Ordered plan +
one-tap deeplinks**

Settings screens, verbatim

Deterministic core

Rules extract, ground and validate every step. The LLM is optional enrichment — never the source of truth.

One-tap deeplinks

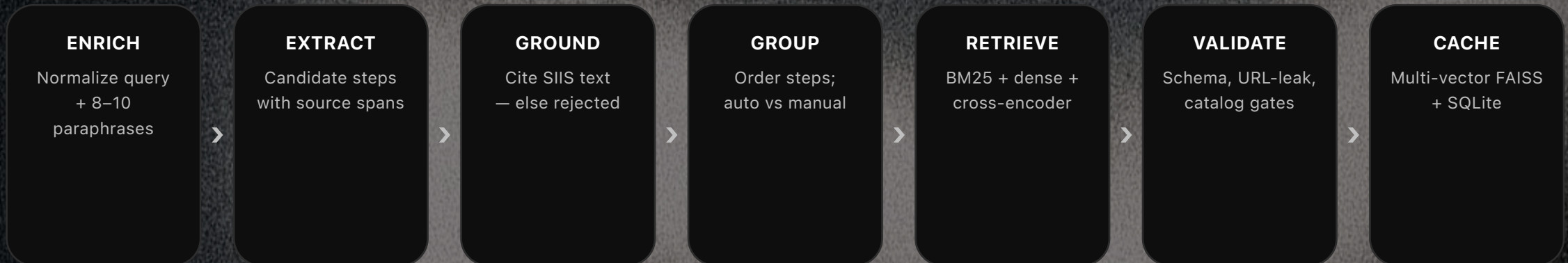
Hybrid retrieval picks the exact Settings screen from 578 catalog entries — copied verbatim, never invented.

Fast by design

A multi-vector paraphrase cache answers repeat complaints in ~40 ms (P95). The API stays open and unauthenticated.

A pipeline that can't hallucinate

Seven stages. Every stage is verifiable — uncited or ungrounded output is rejected, not repaired.



DESIGN RULE

The LLM (Google Gemini) is an optional enrichment layer. With no key — or under rate limits — the pipeline degrades gracefully to pure rules at \$0.00/query.

Boring where it counts. Sharp where it matters.

FastAPI

Open, unauthenticated REST API
GET /health · POST /v1/troubleshoot

BM25 + FAISS

Hybrid retrieval + cross-encoder
rerank for exact deeplinks

Google Gemini

Optional enrichment only —
never required

SQLite

Cached plans +
vector index metadata

Docker

One-command,
reproducible deployment

pytest

91 tests · hard-gated
eval harness

Why this wins

- 01** No hallucinations — by construction. Every step cites a verbatim SITS span. Uncited plans are rejected.
- 02** Verbatim deeplinks. Only catalog entries, copied exactly from 578 Settings screens. Nothing invented.
- 03** Multi-vector paraphrase cache. 8–10 paraphrases embedded up front — 97.5% hit on unseen rewordings, 0% false hits.
- 04** Hybrid retrieval + rerank. Lexical + dense + cross-encoder with leaf-screen preference: 100% deeplink accuracy.
- 05** Graceful LLM degradation. No key or rate-limited? Rules take over — all gates green at \$0.00/query.

Every hard gate passes

20/20 responses fully compliant · results.json ships 8–10 query variations per query

DEEPLINK
ACCURACY

✓ PASS

100%

16 hand-labelled

SCHEMA
VIOLATIONS

✓ PASS

0

20 responses · budget 0

URL LEAKS

✓ PASS

0

in responses · budget 0

FALSE HITS

✓ PASS

0%

on negative queries

CACHE-HIT P95

✓ PASS

~40 ms

budget ≤ 300 ms

NEW-QUERY P95

✓ PASS

~600 ms

budget ≤ 8000 ms

UNSEEN-
PARAPHRASE HIT

✓ PASS

97.5%

cache hit rate

RULES-MODE COST

✓ PASS

\$0.00

per query

Each component earns its place

DEEPLINK RETRIEVAL

Dense-only → Hybrid (BM25 + dense + cross-encoder rerank)

69% → 100%

PARAPHRASE CACHE HIT RATE

Single-vector → Multi-vector (8–10 paraphrases)

48% → 98%

Rules-only pipeline: 20/20 compliant · 0 violations · \$0.00/query — the deterministic core carries the product on its own.

One call. One validated plan.

REQUEST

POST /v1/troubleshoot

```
{
  "query": "My Galaxy screen keeps flickering",
  "siis_response": {
    "title": "Screen Flicker Troubleshooting",
    "body": "If your screen flickers, first verify..."
  }
}
```

RESPONSE

200 OK · 404 ms · \$0.0000

```
{
  "contexts": [{
    "goal": "Follow these steps to perform
              this Screen Flicker Troubleshooting",
    "title": "Screen Flicker Troubleshooting",
    "actions": [{
      "actionName": "Verify Your Phone's
                    Display Settings",
      "category": "auto",
      "actionableDeeplink": {
        "title": "Display",
        "uri": "<verbatim catalog uri>"
      }
    }]
  }]
}
```

Honest about the edges

LIMITATIONS

- LLM enrichment is optional — free-tier rate limits apply
- English-only queries in this version
- Evaluated on the 20 provided SIIS articles
- Deeplink catalog fixed at 578 entries

FUTURE WORK

- Multilingual query support
- Catalog auto-expansion as Settings evolve
- On-device small-model enrichment
- Feedback loop from real resolution outcomes

Thank you.

Theme 2 · Smart Guided Troubleshooting Engine

Repository [https://github.com/armaan-tayal/TIET_Agentic_Alchemists]

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